

Interpretable Rotor-Bar Fault Diagnosis of Squirrel-Cage Induction Motors Based on Polynomial Temporal Convolutional Network

APPENDIX

This supplementary appendix reports validation that are not fully included in the main manuscript because of space limitations. All experiments follow the same diagnostic task as the manuscript: SCIM rotor-bar porosity and broken-bar defect diagnosis using the phase-C open-circuit induced-voltage response. The dataset is constructed from controlled rotor specimens and is intended for a controlled pre-assembly rotor-quality inspection scenario. The appendix contains four groups of supplementary materials:

- 1) Additional diagnostic-reliability evidence, including K-fold validation, convergence curves, and repeated random-seed trials.
- 2) Structural comparison between diagonal polynomial form and full local Volterra form, to justify the adopted model simplification.
- 3) Polynomial-order selection and noise sensitivity, to explain the chosen model configuration and its robustness limits.
- 4) Hyperparameter configuration, to provide complete implementation transparency.

The measured diagnostic signal is the phase-C open-circuit induced-voltage response, rather than conventional three-phase stator current or rotor-bar current. The dataset is constructed from controlled rotor specimens and should be interpreted as a controlled pre-assembly rotor-quality inspection benchmark.

A. Supplementary Validation for Diagnostic Accuracy

To assess whether the reported performance depends on a favorable data partition, K-fold validation was conducted using the same non-overlapping signal-construction principle. As shown in Table A-I, the proposed PTCN achieved $99.78 \pm 0.30\%$ accuracy, with precision, recall, and F1-score remaining at comparable levels. This result indicates that the diagnostic performance is stable across folds rather than being caused by a single split.

To further evaluate stochastic training stability, four random seeds were used, and each seed was repeated five times under the same network configuration, optimizer, batch size, dataset partitioning, and evaluation protocol. Table A-II reports 20 independent training-testing trials. The overall accuracy of $99.87 \pm 0.41\%$ provides a more conservative and representative estimate than a single-run result.

Fig. A1 provides supplementary convergence evidence. The accuracy increases rapidly and approaches saturation, while the loss decreases smoothly without evident late-stage oscillation. This behavior is consistent with stable optimization rather than delayed memorization or unstable stochastic convergence.

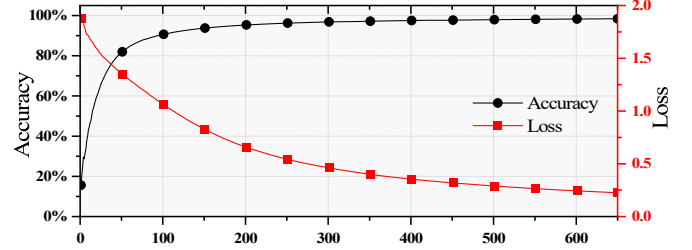


Fig. A1. Training accuracy and loss curves of the proposed PTCN.

TABLE A-I.
K-FOLD VALIDATION RESULTS OF THE PROPOSED PTCN

Fold	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Fold 1	100.00	100.00	100.00	100.00
Fold 2	100.00	100.00	100.00	100.00
Fold 3	99.41	99.44	99.41	99.41
Fold 4	100.00	100.00	100.00	100.00
Fold 5	99.48	99.50	99.48	99.48
Mean $\pm$ Std	99.78 ± 0.30	99.79 ± 0.29	99.78 ± 0.30	99.78 ± 0.30

TABLE A-II.
REPEATED RANDOM-SEED TRAINING RESULTS OF THE PROPOSED PTCN.

Random seed	Accuracy (%)
42	99.95 ± 0.11
61	99.94 ± 0.12
88	99.63 ± 0.81
123	99.94 ± 0.11
Overall	99.87 ± 0.41

B. Diagonal Polynomial Form vs. Full Volterra Form

A controlled comparison was conducted by replacing the diagonal PolyConv operator with a full local Volterra expansion while keeping the downstream architecture and training settings unchanged. The full local Volterra form includes cross terms among local samples, whereas the diagonal form retains same-lag self-product terms. Therefore, this experiment evaluates whether the additional cross terms improve diagnostic performance under the available data and training budget.

As shown in Table A-III, the diagonal form outperforms the full local Volterra form under clean and noisy conditions. The result suggests that the additional cross terms do not provide useful diagnostic gain in this task and may instead increase optimization difficulty, noise sensitivity, and overfitting risk.

Table A-IV further shows that the diagonal form preserves clearer order-wise associations between polynomial energy components and waveform-level statistical descriptors. This supports the use of a structured diagonal polynomial basis as a

sparse, interpretable feature operator, rather than treating PolyConv as a complete Volterra-series implementation

TABLE A-III.

ACCURACY COMPARISON OF DIAGONAL AND FULL VOLTERRA FORMS.

Model	Clean (%)	SNR20 (%)	SNR10 (%)
Diagonal Volterra form	99.87	99.85	92.60
Full Volterra form	71.42	51.16	26.14

TABLE A-IV.

COMPARISON OF INTERPRETABILITY-RELATED MOMENT-ENERGY CORRELATIONS BETWEEN DIAGONAL AND FULL VOLTERRA FORMS.

Model	E1-Mean	E2-Variance	E3-Skewness	E4-Kurtosis
Diagonal Volterra form	0.638	0.785	0.938	0.952
Full Volterra form	0.060	-0.801	0.445	0.912

C. Polynomial-Order Selection and Noise Sensitivity

In addition to the preliminary order-selection experiment, we further designed a noise sensitivity index (NSI) to quantitatively evaluate the perturbation sensitivity of each polynomial-order component. The NSI of the p^{th} order component is defined as:

$$\text{NSI}_p = \frac{|\Phi_p(x + \eta) - \Phi_p(x)|_F}{|\Phi_p(x)|_F + \varepsilon} \quad (\text{A-1})$$

where x denotes the original input signal, η denotes the injected Gaussian noise, $\Phi_p(\cdot)$ denotes the p^{th} order PolyConv feature response, $|\cdot|_F$ is the Frobenius norm, and ε is a small constant used for numerical stability. A larger NSI value indicates that the corresponding polynomial-order response is more sensitive to noise perturbation. The NSI results are summarized in Table A-V.

An isolated PolyConv classifier was used to compare different maximum polynomial orders. As summarized in Table A-VI, the fourth-order configuration provides the best overall balance across clean, SNR20, SNR15, and SNR10 conditions. Lower-order settings are less expressive, whereas the fifth-order setting becomes unstable under stronger noise

D. Hyperparameter Settings

The hyperparameters used for the supplementary experiments are summarized in Table A-VII. These settings are kept consistent with the main manuscript unless otherwise stated.

TABLE A-V.

NOISE SENSITIVITY INDEX OF DIFFERENT POLYNOMIAL ORDERS.

Order	NSI
1 st order	0.6188 ± 0.0526
2 nd order	0.6121 ± 0.0647
3 rd order	1.3306 ± 0.0709
4 th order	1.4886 ± 0.1655
5 th order	2.3239 ± 0.2211

TABLE A-VI.

PRELIMINARY ORDER-SELECTION RESULTS OF THE POLYCONV LAYER

Order	Clean (%)	SNR20 (%)	SNR15 (%)	SNR10 (%)
1 st order	89.29 ± 8.08	82.88 ± 4.42	60.22 ± 1.73	38.77 ± 2.83
2 nd order	84.29 ± 12.19	85.58 ± 7.80	62.24 ± 4.89	35.95 ± 3.84
3 rd order	89.07 ± 5.78	85.93 ± 5.17	55.21 ± 7.45	31.34 ± 6.27
4 th order	93.29 ± 4.61	88.65 ± 6.00	79.46 ± 6.53	39.43 ± 1.32
5 th order	85.00 ± 12.11	77.69 ± 8.05	53.60 ± 5.20	20.64 ± 3.76

TABLE A-VII.

HYPERPARAMETER SETTINGS OF PROPOSED MODELS

Category	Hyperparameter	Value	Category	Hyperparameter	Value
Training	Input size	256×1	Training	Optimizer	Adam
	Number of classes	7		Learning rate	0.001
	Iteration	655		Loss function	Categorical cross-entropy
	Batch size	32		Random seed	42
PolyConv	Maximum order	4	PolyConv	Padding	Same
	Polynomial type	Diagonal		activation	ReLU
	Number of filters	32		Kernel size / stride	4 / 4
Constraints	λ_{orth}	0.02	Downstream Network	TCN dilation rates	1, 2, 4
	λ_{energy}	0.10		TCN kernel size	4
	λ_{smooth}	0.001		Classifier hidden units	128
Downstream Network	Classifier L2	0.1	Downstream Network	Attention heads	8
	Dropout	0.5		Attention key dimension	16